



Tribhuvan University Institute of Engineering Khwopa College of Engineering

A Mid Term Defense Presentation on NSL RECOGNITION FOR EMERGENCY PHRASES:

A Landmark-Based Real-Time Recognition System using MediaPipe and BiLSTM

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Presentation Outlines

- 1. Introduction**
- 2. Motivation & Objective**
- 3. Scope**
- 4. Literature Review**
- 5. Methodology**
- 6. Result**
- 7. Conclusion**

1. Introduction

- Sign language: Deaf communication
- Emergencies need instant understanding
- We target six emergency phrases
- Webcam reads hands and body
- Focus on Nepali Sign Language

2. Motivation & Objective

- Deaf face barriers in emergencies
- Nepali Sign Language is under-resourced
- Build reliable landmark data pipeline
- Collect dataset across multiple signers
- Prepare data for sequence model
- Keep design browser-portable

3. Scope

- **Six fixed emergency phrases** + 1 “none” class
- One sign per clip
- **Landmarks** stored, not video
- Webcam input, runs on **CPU**
- Current phase: **Data pipeline**

4. Literature Review

Study	Method	Gap	Q
Shrestha et al., 2024	MediaPipe + CNN on skeleton images	Static only, data leakage	
Poudel et al., 2025	MobileNetV2 / ResNet50	Static only, no feedback	
Ligal & Baral, 2022	CNN + RNN / Vision Transformer	Limited coverage	
Mali et al., 2018	CNN + red gloves required	Impractical, tiny vocabulary	
NSL23 dataset, 2024	First public video NSL dataset	No recognition system built on it	



4. Literature Review

Ligal & Baral (2022) – Nepali Sign Language Gesture Recognition Using Deep Learning

- This paper explores deep architectures for NSL gesture recognition.
- One variant combines a CNN with a recurrent neural network.
- Another variant pairs a CNN with a Vision Transformer instead.
- Both follow the data- and compute-heavy pixel-based formulation.
- Confirms NSL research interest, but targets neither emergencies nor real-time use.



4. Literature Review

Goyal & Velmathi (2023) – Indian Sign Language Recognition Using MediaPipe

Holistic

- This paper applies MediaPipe Holistic to Indian Sign Language recognition.
- Body pose and both hands are tracked together in every frame.
- LSTMs are used for dynamic signs and CNNs for static ones.
- It validates the landmark-plus-temporal-model direction adopted here.
- Closest method to ours, but targets ISL and gives no audio output.



4. Literature Review

Sunuwar, Borah & Kharga (2024) – NSL23 Dataset for Alphabets of Nepali Sign Language

- This paper releases NSL23, the first publicly available NSL dataset.
- It contains 630 videos covering the 49 NSL alphabet characters.
- Fourteen volunteers performed the signs for these recordings.
- Video is used because some NSL signs are defined by motion.
- Proves NSL data collection is feasible, but covers the alphabet only.



4. Literature Review

Shrestha, Thing, Dhakal, Dahal, Karki & Guragai (2024) – Nepali Sign Language

Letter Detection and Finger Spelling Using MediaPipe and CNN

- This paper detects NSL letters using MediaPipe hand landmarks.
- The landmarks are re-rendered as skeleton images for a CNN.
- High accuracy is reported on the fingerspelling alphabet.
- Training and test signers are not separated during evaluation.
- Closest prior NSL landmark work, but static and not signer-independent.



4. Literature Review

Poudel et al. (2025) – Nepali Sign Language Characters Recognition: Dataset

Development and Deep Learning Approaches

- This paper introduces a 36-class static NSL image dataset.
- Fine-tuned MobileNetV2 and ResNet50 networks are evaluated on it.
- Both models reach roughly 90 percent recognition accuracy.
- The formulation remains single-image and pixel-based throughout.
- Recent NSL benchmark, but excludes the motion-based phrase signs.

5. Methodology



#	Organisation / School	Type	Address / Location	District	Phone
1	Bhaktapur Association of the Deaf	Deaf association (district)	Thali, Ward 10, Bhaktapur	Bhaktapur	Thali, Ward 10, Bhaktapur
2	RCRD & CBR Bhaktapur (Community Based Rehabilitation Org.)	Disability / rehab NGO	MCFG+7FV, Bhaktapur	Bhaktapur	+977 1-6612630
3	Satprayas Special Needs School Nepal	Special-needs school	Bhimsenthan-5, Bhaktapur	Bhaktapur	Not listed on Maps - visit / message on Facebook
4	National Federation of the Deaf Nepal (NFDN)	National apex body	Ranibari Marg, Kathmandu	Kathmandu	+977 1-4220514 / +977 1-4535515
5	Kathmandu Association of the Deaf (KAD)	Deaf association	Gairidhara Sadak, Kathmandu	Kathmandu	+977 1-4423256
6	Central Secondary/Higher Secondary School for the Deaf (Naxal School for the Deaf)	Deaf school (ECD-12 / Bachelor)	Balmandir, Naxal, Kathmandu	Kathmandu	+977 1-4410179
7	Nepal Deaf Art & Culture Society	Deaf cultural NGO	Kathmandu (north)	Kathmandu	+977 985-1237896
8	Nepal Association of the Hard of Hearing	Association / support org	Kathmandu (Kalanki-Sanepa area)	Kathmandu	+977 1-4330921
9	Patan Community Based Rehabilitation Organization	Disability / rehab NGO	Shankhamul Sadak, Lalitpur	Lalitpur	+977 1-5268291
10	Kavre Deaf School (Deaf School Banepa)	Deaf school	JGP9+3XX, Banepa	Kavrepalanchok	+977 984-1300823
11	Shree Padma Secondary School (deaf/special unit)	Mainstream school with special unit	MCFG+3WW, Bhaktapur	Bhaktapur	+977 1-6610326
12	Sindhupalchok Association of the Deaf (Deaf School, Barhabise)	Deaf school / association	Barhabise, Sindhupalchok	Sindhupalchok	+977 11-63032

RCRD & CBR Bhaktapur



National Federation of the Deaf Nepal (NFDN)





Khwopa College of Engineering



NSL Landmark Recorder

Output folder

M:/Projects/minor-project/Data Browse...

Signer ID

signer03

Phrase (class)

☒ 1. I can't

☐ 2. The bi

☐ 3. Call th

☐ 4. I need

☐ 5. Help r

☐ 6. I need

☐ 7. Unkn

Start reco

Restored las

Clips on dis

cal_police:

need_toilet:

NSL Landmark Recorder

Output folder

Browse...

Signer ID

signer01

Phrase (class)

☐ 1. I can't breathe (Medical)

☐ 2. The building is on fire (Fire)

☐ 3. Call the police (Crime)

☒ 4. I need an ambulance (Medical)

☐ 5. Help me / I am in danger (Generic)

☐ 6. I need to go to the toilet (Basic need)

☐ 7. Unknown / none of the above (negatives)

Start recording

Stop & save

Discard

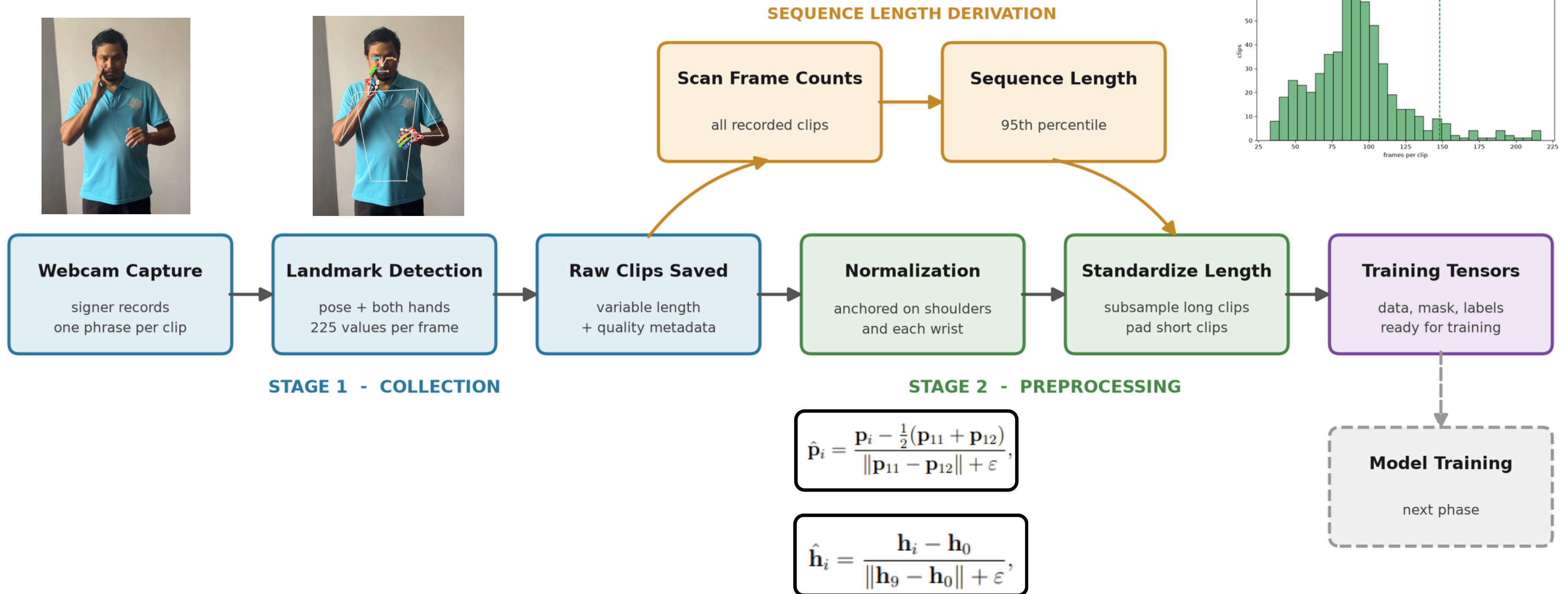
Choose an output folder to begin.

Outcomes:

- Complete dataset successfully built
- Seven phrases(6 emergency, 1 “none”)
- 570 usable clips total

5. Methodology

Methodology - Data Collection & Preprocessing Pipeline



6. RESULT

- All clips normalized and standardized
- Training tensors generated
- Data ready for model training

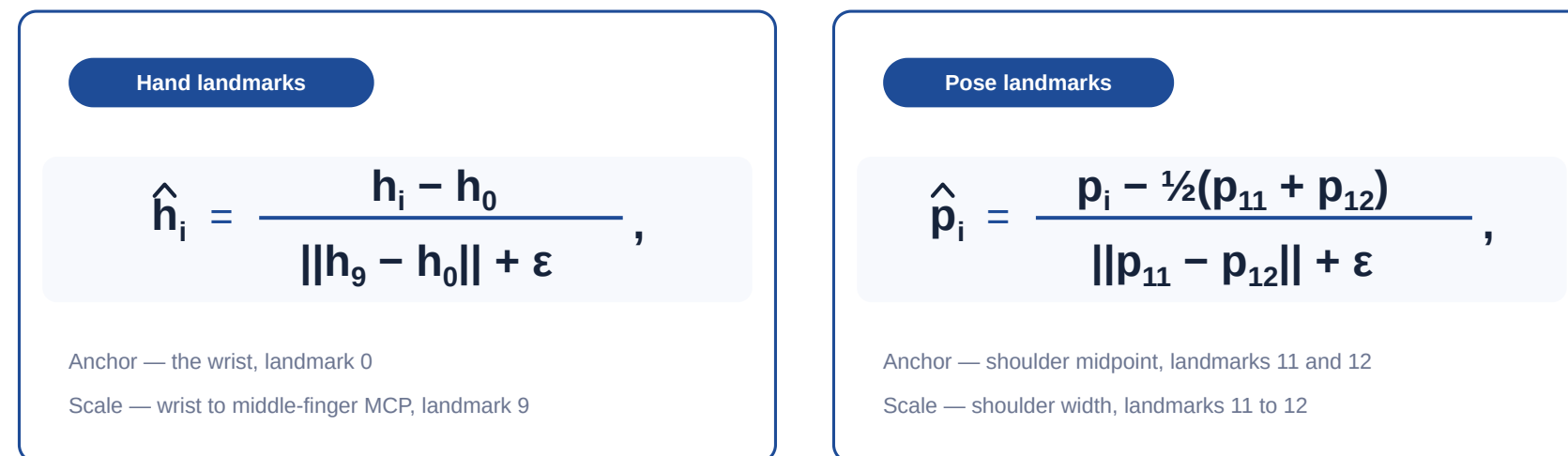
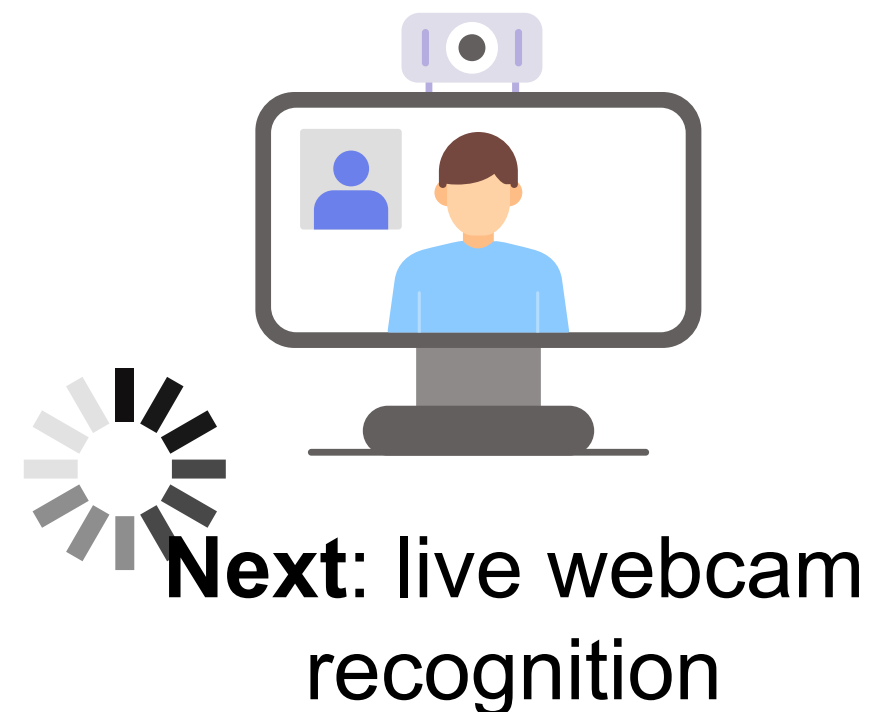
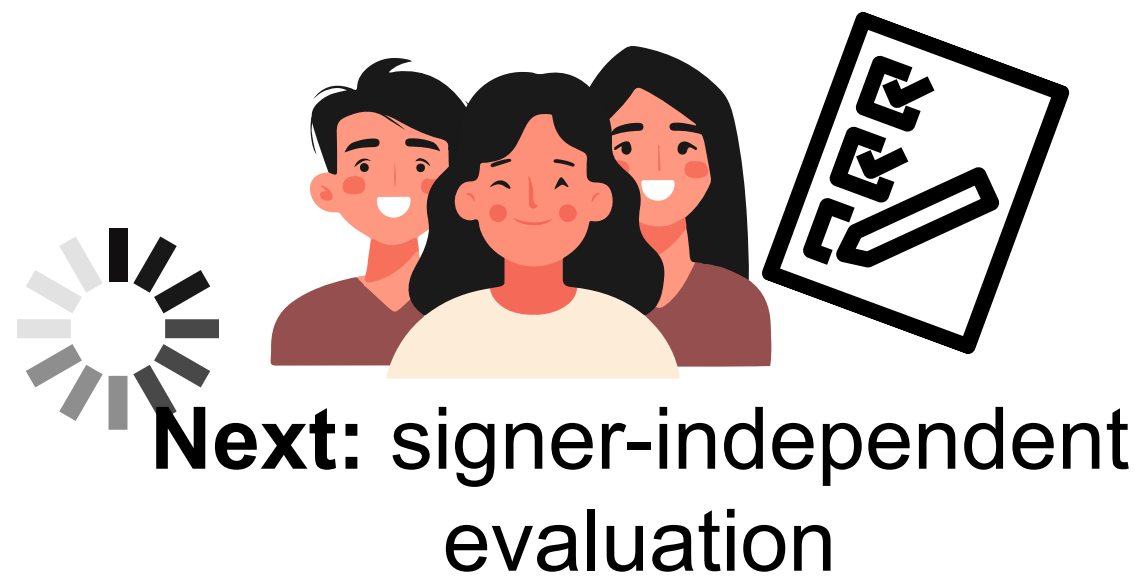
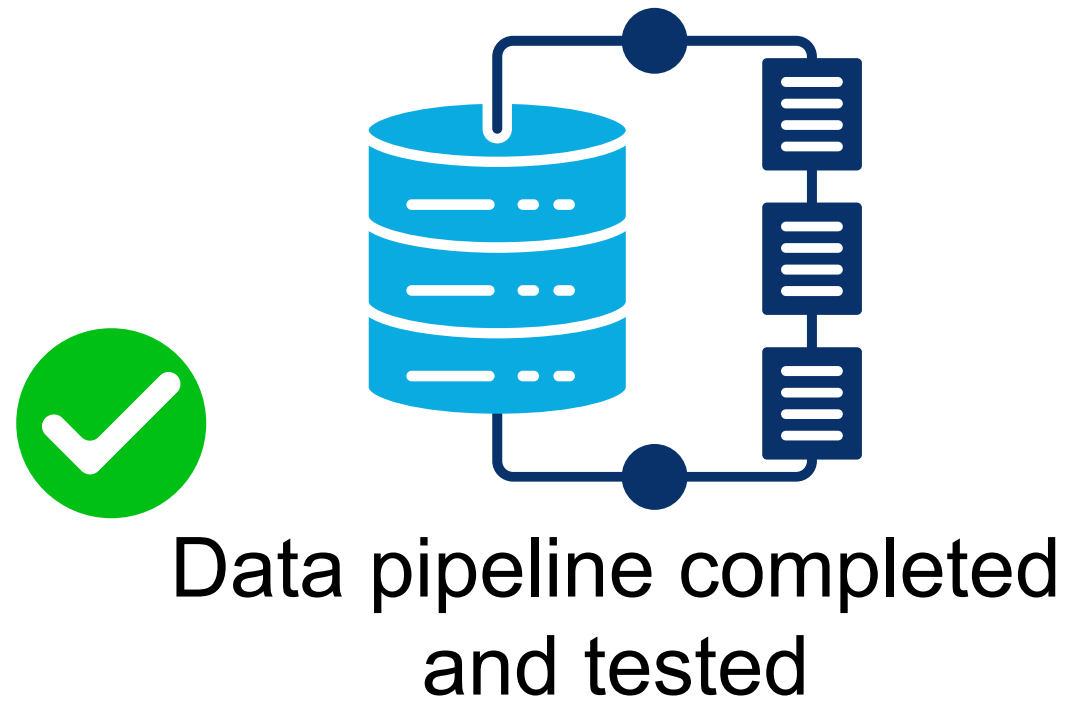


fig: Dual Anchor Normalization
formulae



fig: Training Tensors

7. CONCLUSION AND FUTURE WORK



THANK YOU!